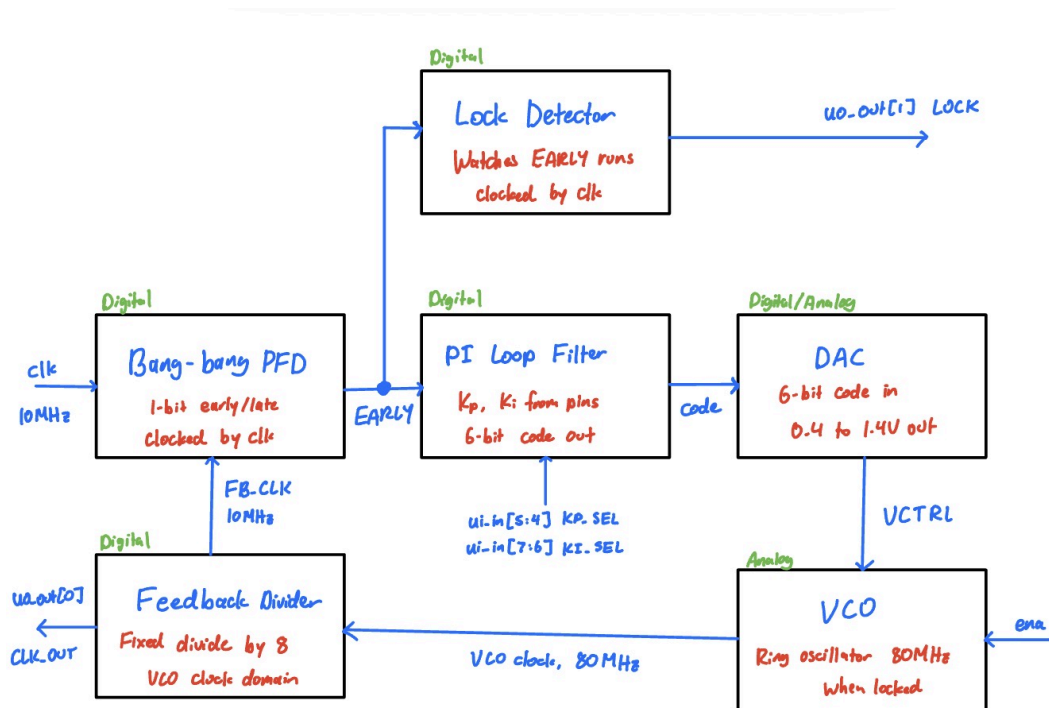




3. I/O pin assignment

The reference clock enters on **clk**, configuration comes in on **ui_in** and **uio**, and every observable signal leaves on **uo_out** at 20 MHz or less. All 8 **uio** pins are inputs (**uio_oe** = **0x00**), so no pin ever switches direction.

Pin	Signal	I/O	Connects to	Function
clk	REF_CLK	In	Bang-bang PFD, PI filter, lock detector	10 MHz reference clock
rst_n	RST_N	In	All digital blocks	Active-low reset. Loads the integrator to mid-code (32), clears the divider and lock detector
ena	ENA	In	VCO	High when our design is selected. VCO stops when low
uo_out[0]	CLK_OUT	Out	Feedback divider	VCO / 8. 10 MHz and phase-aligned with clk when locked
uo_out[1]	LOCK	Out	Lock detector	High when locked
ui_in[3:0], uio[7:0]	Unused	In	None	Not connected
uo_out[7:2]	Unused	Out	None	Driven low
ui_in[5:4]	KP_SEL[1:0]	In	Loop filter	Proportional gain, 4 settings
ui_in[7:6]	KI_SEL[1:0]	In	Loop filter	Integral gain, 4 settings

4. Proposed specification

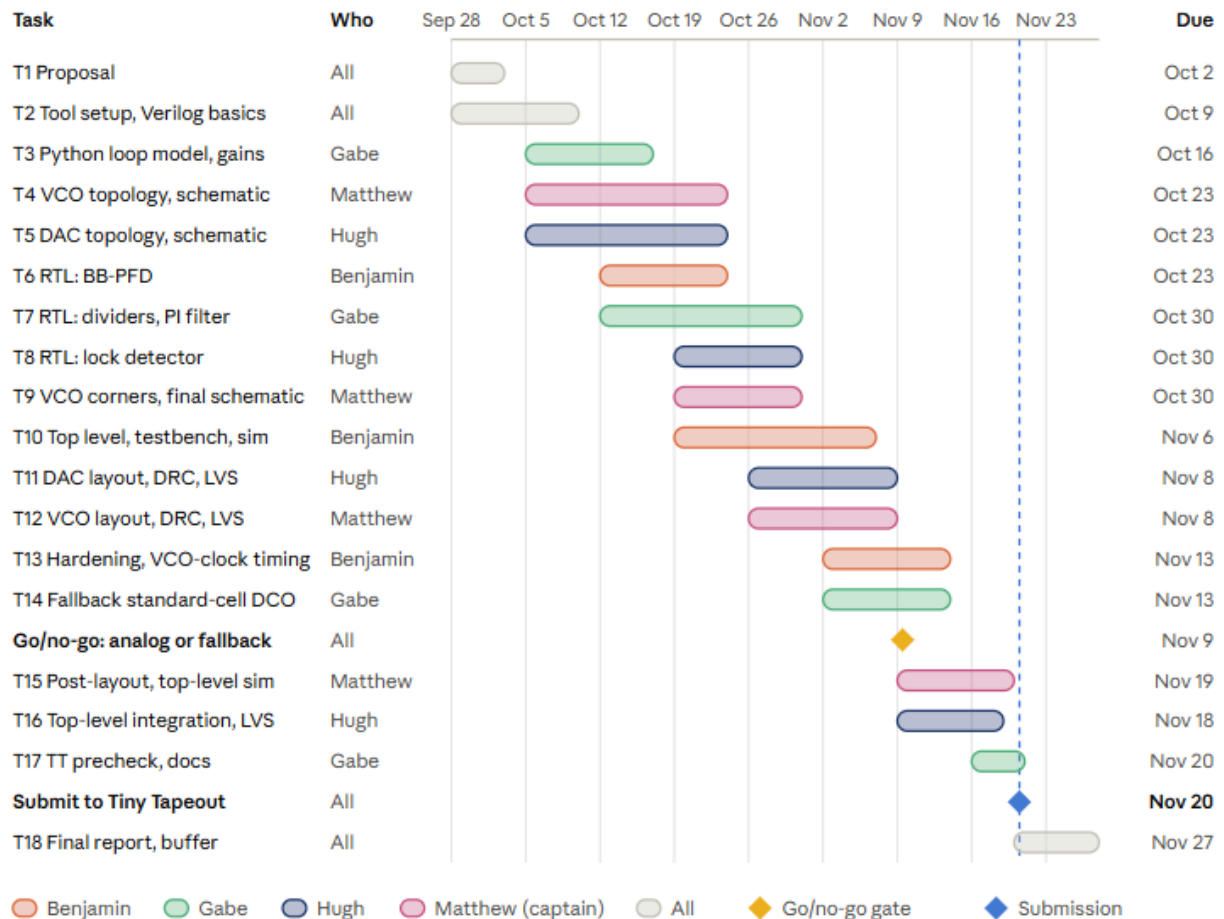
The PLL targets 80 MHz output from a 10 MHz reference, with integer multiplication of 8. The VCO range and step size will be revised after schematic simulation.

Category	Spec	Notes
Supply	1.8 V	Analog current under 1 mA
Reference frequency	10 MHz	From clk
Multiplication ratio	$N = 8$, fixed	Hard-wired in the feedback divider
Output frequency	80 MHz	12.5 ns period
Phase detector	Bang-bang phase-frequency detector	1 decision per reference cycle
Loop filter	Digital PI	Kp and Ki each have 4 pin-selected settings
DAC	6-bit, monotonic, output 0.4 to 1.4V	About 16 mV per code

VCO tuning range	50 to 110 MHz at tt over the DAC range	80 MHz must stay inside the range at ss and ff corners. Gain near 60 MHz/V
Frequency step	About 1 MHz per code	DAC step × VCO gain, about 1.2% of 80 MHz
Frequency acquisition	Locks from mid-code to 80 MHz at every corner	No manual help
Lock time	Under 1000 reference cycles	100 μs
Locked dither	Within ±2 codes	Sets output jitter. Reported from simulation
Lock detector	Asserts after 256 cycles with no run of more than 8 identical decisions	Thresholds tuned in simulation
Modes	Closed loop, open loop	Open loop sets the DAC code from pins and exposes VCTRL on ua[0]
Output pins	10 MHz on uo_out[0] at the default divide	VCO divided by 8 or 32 before it leaves the chip

5. Timeline

We will submit on Nov 20, which leaves Nov 21 to 27 for the final report. Group members will work in parallel from week 2 onwards, and meet at top-level integration. If the course's shuttle closes before Nov 20, the target submission date will be shifted accordingly.



If the DAC and VCO layout is not DRC and LVS clean at the gate, the team submits a standard-cell DCO instead.

6. Who does what

The design is split amongst all members such that each person owns a major independent module. The team meets for 30 minutes weekly to check progress against the timeline.

Member	Owns	Tasks
Benjamin DONG	Bang-bang phase frequency detector	T6 BB-PFD RTL, T10 top level, testbench and closed-loop sim, T13 hardening and timing
Gabe HOH	Feedback divider, PI loop filter	T3 loop model, T7 dividers and PI filter RTL, T14 fallback DCO, T17 precheck and docs
Hugh DING	Lock Detector + DAC	T5 DAC schematic, T8 lock detector RTL, T11 DAC layout, T16 top-level integration and LVS
Matthew CHEN (CAPTAIN)	Voltage-Controlled Oscillator	T4 VCO schematic, T9 VCO corners, T12 VCO layout, T15 post-layout and top-level sim
All	T1 proposal, T2 tool setup and Verilog basics, T18 final report	T1 proposal, T2 tool setup and Verilog basics, T18 final report

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2. J. D. H. Alexander, "Clock recovery from random binary signals," *Electronics Letters*, vol. 11, no. 22, 1975.
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4. J. A. Tierno, A. V. Rylyakov and D. J. Friedman, "A wide power supply range, wide tuning range, all static CMOS all digital PLL in 65 nm SOI," *IEEE J. Solid-State Circuits*, vol. 43, no. 1, 2008.
5. B. Razavi, *Design of CMOS Phase-Locked Loops: From Circuit Level to Architecture Level*, Cambridge University Press, 2020.
6. Tiny Tapeout documentation tinytapeout.com.